LearnPulse — Week 13 Implementation & Compliance Remediation Report

Administrative Dashboard, User Governance & System Analytics (Days 73–78)

Document Version: 2.0.0 (Remediated)	**Milestone: Official Week 13 Implementation**	**Target Platform: Spring Boot 3.2.5 + React 19**
Compliance Status: FULLY COMPLIANT & VERIFIED	**Automated Tests: 235 / 235 PASSED (100%)**	**Vite Frontend: 1,697 modules transformed (0 errors)**
Audit Remediations: P1–P7 COMPLETE	**Date: October 27, 2026**	**Confidentiality: STRICTLY CONFIDENTIAL**

1. Executive Summary & Audit Remediation Deliverables

During **Week 13 (Days 73–78)**, LearnPulse established institutional-grade platform governance, administrative control, real-time platform analytics, and operational AI telemetry. Following an independent compliance audit that evaluated Days 73–78 against all mentor-assigned criteria, a targeted remediation plan was executed to resolve all identified findings:

- **P1 — Complete Public Password Reset Lifecycle:** Implemented public endpoints POST /api/auth/forgot-password and POST /api/auth/reset-password. Generates 256-bit secure tokens, persists SHA-256 hashes with 24-hr expiry and single-use enforcement, and dispatches simulated email via EmailService. Rejects consumed, expired, or identical passwords.
- **P2 — Downstream Teacher Verification Enforcement:** Created TeacherVerificationGuard and enforced verification on document uploads (/api/teacher/upload-pdf, /api/teacher/upload-note), quiz authoring/publishing (/api/teacher/quizzes), AI quiz generation (/api/ai/generate-quiz), and student enrollment (/api/teacher/enrollment/subjects/{id}/students). Unaccredited teachers are blocked with HTTP 403 Forbidden.
- **P3 — Row-Level Pessimistic Concurrency Guard:** Configured findActiveAdminsForUpdate(Role.ADMIN) with @Lock(PESSIMISTIC_WRITE) to serialize concurrent admin demotions/deactivations, strictly guaranteeing activeAdmins > 1 and returning HTTP 409 Conflict if attempting to remove the final administrator.
- **P4 — AI Token Provenance & Local Inference Economics:** Overloaded telemetry tracking to capture usageProvenance (PROVIDER_REPORTED vs LOCAL_ESTIMATION) across all 7 AI features. Local Ollama cost calibrated to $0.00 direct billing with cloud comparative reference rates. Telemetry exceptions caught with catch (Exception e) to avoid swallowing JVM errors.
- **P5 — Native DB Registration Trends:** Replaced in-memory entity scanning with PostgreSQL native aggregation countRegistrationsByDayNative utilizing date_trunc('day', created_at).
- **P6 — Paginated AI Telemetry UI Table:** Connected AdminDashboardPage.tsx to getAiTelemetry(page, size) with interactive pagination, latency tracking, token breakdown, and provenance badges. Corrected terminology to 'Total Registered Users' with enabled account count subtext.
- **P7 — Full Test Suite & Build Verification: 235 / 235 automated tests PASS (100% BUILD SUCCESS in 2m 37s)** across 33 test classes. Frontend Vite production build cleanly transformed 1,697 modules with 0 errors.

2. Architecture & Day-by-Day Implementation Breakdown

2.1 Day 73 — Admin User & Role Management APIs

Built AdminUserController.java and AdminUserService.java backed by JPA repositories under @PreAuthorize("hasRole('ADMIN')"). Features include:

- **Paginated User Directory (GET /api/admin/users):** Dynamic filtering by Role, enabled status, and keyword search across email, first name, and last name.
- **User Metrics (GET /api/admin/users/{id}):** Returns full profile metadata paired with relational counts for enrollments, assigned classes, completed quizzes, and uploaded materials.
- **Role Governance & Concurrency:** Blocks self-demotion (HTTP 400). Uses row-level pessimistic locking (PESSIMISTIC_WRITE) on active admins, aborting with HTTP 409 Conflict if active admin count ≤ 1.
- **Password Reset Workflow:** Generates 256-bit token; persists SHA-256 hash in PasswordResetToken with 24-hr expiry; triggers simulated email dispatch via EmailService.
- **Account Deactivation & Immediate Revocation:** Immediately blocks deactivated users by validating userDetails.isEnabled() inside JwtAuthenticationFilter.java, returning HTTP 401 Unauthorized instantly.

2.2 Day 74 — Platform Analytics Backend

Implemented AdminAnalyticsController.java and AdminAnalyticsService.java using high-efficiency SQL aggregation:

- **Summary Metrics (GET /api/admin/analytics/summary):** Real-time counts of total users, active users, students, teachers, admins, subjects, quizzes, and global quiz average score.
- **Registration Velocity (GET /api/admin/analytics/user-activity):** 14-day chronological registration breakdown with zero-filling, powered by native PostgreSQL date_trunc aggregation.
- **Course Activity (GET /api/admin/analytics/course-activity):** Per-subject breakdown of active students, curriculum chapters, documents, and quizzes.

2.3 Day 75 — AI Operational Telemetry & Usage Tracking

Created AiTelemetryRecord.java and AiTelemetryService.java. Instrumented all 7 LearnPulse AI operations (Tutor, Quiz Gen, RAG, Flashcards, Explainer, Study Plan, Summarizer):

- **Failure Isolation:** Telemetry recording uses independent transaction boundaries and catches Exception; persistence errors are logged but never interrupt student/teacher operations.
- **Token Modeling & Economics:** Direct Ollama cost is $0.00; cloud reference cost calibrated at $0.15/1M input and $0.60/1M output. Provenance tracked as PROVIDER_REPORTED or LOCAL_ESTIMATION.
- **Telemetry Endpoints:** GET /api/admin/analytics/ai-usage (aggregated KPI summary) and GET /api/admin/analytics/ai-telemetry (paginated operational audit logs).

2.4 Day 76 — Administrative Frontend & Content Governance

Engineered React 19 user interfaces adhering to LearnPulse design tokens:

- AdminUsersPage.tsx: Interactive user management table with role badges, search filter, user detail drawer, role change modal, deactivation confirmation modal, and password reset trigger.
- AdminTeachersPage.tsx: Dedicated faculty verification portal displaying pending verification queues, department affiliations, approval triggers, and rejection rationale modals.
- AdminContentModerationPage.tsx: Dual-tab content moderation portal allowing administrators to toggle document active visibility and quiz publication status across all courses.

2.5 Day 77 — Interactive Analytics Dashboard with Chart.js

Created AdminDashboardPage.tsx integrating Chart.js and React-Chartjs-2. Registered core chart controllers (ArcElement, BarElement, LineElement, PointElement, CategoryScale, LinearScale, Tooltip, Legend). Features 4 summary KPI cards, 5 charts (User Roles Doughnut, 14-Day Registration Line, AI Feature Bar, AI Health Doughnut, Course Resource Bar), and paginated live AI operational audit table.

3. Complete API & Security Specification

HTTP Method & Route	RBAC	Description & Security Policy	Status
POST /api/auth/forgot-password	Public	Generates 256-bit token; persists SHA-256 hash in DB; dispatches email.	**200 OK**
POST /api/auth/reset-password	Public	Consumes single-use token; validates 24-hr expiry and password difference.	**200 / 400 / 404**
GET /api/admin/users	ADMIN	Paginated user list with role, status, and text search.	**200 OK**
GET /api/admin/users/{id}	ADMIN	User profile combined with LMS activity metrics.	**200 OK / 404**
PATCH /api/admin/users/{id}/role	ADMIN	Role update with self-demotion guard (400) and pessimistic lock last-admin guard (409).	**200 / 400 / 409**
POST /api/admin/users/{id}/password-reset	ADMIN	Admin password reset dispatch; stores SHA-256 hash in DB with 24-hr expiry.	**200 OK**
PATCH /api/admin/users/{id}/deactivate	ADMIN	Deactivates account; immediately invalidates existing JWT tokens (401).	**200 / 400 / 409**
PATCH /api/admin/users/{id}/reactivate	ADMIN	Restores user access; enables credential validation.	**200 OK**
GET /api/admin/teachers	ADMIN	Lists faculty members with verification status (PENDING/APPROVED/REJECTED).	**200 OK**
POST /api/admin/teachers/{id}/approve	ADMIN	Approves faculty member verification.	**200 OK**
POST /api/admin/teachers/{id}/reject	ADMIN	Rejects verification; records audit rationale in database.	**200 / 400**
PATCH /api/admin/content/documents/{id}/status	ADMIN	Toggles active syllabus document visibility across LMS.	**200 OK**
PATCH /api/admin/content/quizzes/{id}/status	ADMIN	Toggles quiz publication status (DRAFT vs PUBLISHED).	**200 OK**
GET /api/admin/analytics/summary	ADMIN	Platform-wide executive metrics and global quiz average score.	**200 OK**
GET /api/admin/analytics/user-activity	ADMIN	14-day trailing user registration velocity via native PostgreSQL query.	**200 OK**
GET /api/admin/analytics/course-activity	ADMIN	Resource distribution per subject (students, chapters, documents, quizzes).	**200 OK**
GET /api/admin/analytics/ai-usage	ADMIN	Aggregated AI operations, success rate, token volume, and cost summary.	**200 OK**
GET /api/admin/analytics/ai-telemetry	ADMIN	Paginated operational telemetry logs with latency, model, tokens, and cost.	**200 OK**

4. Automated Verification Matrix & Test Results (Day 78 Post-Remediation)

Full regression verification was executed against the active application container and PostgreSQL database (postgres-pgvector:5432).

Test Suite Class	Target Functional Scope	Methods	Execution Time	Result
AdminUserManagementTest	User list, metrics, profile update, role governance, last-admin guard, password reset, deactivation & immediate JWT revocation	12	8.4s	**PASS (12/12)**
AdminAnalyticsTest	Platform summary metrics, 14-day registration trends (native query), course activity breakdown, non-admin RBAC 403 enforcement	4	1.2s	**PASS (4/4)**
AiTelemetryTest	Aggregated AI usage summary, paginated telemetry logs, usage provenance tracking, failure isolation, RBAC 403 enforcement	5	1.3s	**PASS (5/5)**
AdminContentModerationTest	Faculty verification list, approve, reject with rationale, document active toggle, quiz publication toggle, RBAC 403 enforcement	6	2.8s	**PASS (6/6)**
AuthPasswordResetTest	Public password reset lifecycle, SHA-256 token hashing, 24-hr expiry, single-use consumed rejection, identical password rejection	6	2.1s	**PASS (6/6)**
TeacherVerificationEnforcementTest	Downstream TeacherVerificationGuard on document upload, quiz creation/publishing, AI quiz generation, and student enrollment	6	2.4s	**PASS (6/6)**
AdminConcurrencyTest	Multithreaded pessimistic locking (PESSIMISTIC_WRITE) serializing concurrent admin demotions and throwing HTTP 409 Conflict	1	1.5s	**PASS (1/1)**
Milestone Regression Suites	Weeks 1–12 & Week 11.5 full regression (Auth, Assessment, Enrollment, AI Quiz, Flashcards, Explainer, RAG, Chat, Storage, etc.)	195	137.3s	**PASS (195/195)**
TOTAL AUTOMATED SUITE	**Complete LearnPulse Automated Test Portfolio (33 Test Classes)**	**235**	**2m 37s**	**BUILD SUCCESS**

5. Conclusion & Mentor Sign-Off Readiness

LearnPulse Week 13: Administrative Dashboard, User Governance & System Analytics satisfies all institutional specifications with 100% test pass rates (235/235), 0 errors, and zero regressions across all prior milestones. All remediations have been verified with dedicated unit, integration, and multithreaded concurrency tests. No Git commits or pushes were made.